

SARIKELLA MADHU

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OBJECTIVE

Final-year Information Technology undergraduate with a strong foundation in Data Structures, DBMS, Operating Systems, and Computer Networks, and hands-on experience building and deploying full-stack web applications, REST APIs, and LLM-backed retrieval systems. Seeking to apply strong problem-solving and analytical skills to real-world engineering and business operations.

EDUCATION

Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad <i>BE in Information Technology</i>	2023 — 2027 (Expected) CGPA: 9.02/10.0
Sri Chaitanya Junior College, Kukatpally <i>Higher Secondary Education</i>	2021 — 2023 979/1000

PROJECTS

StayDesk — Full-Stack Hostel Operations Platform | *React, Node.js, Express, MySQL, JWT*

- Built a role-based hostel platform for three user roles (student, warden, admin), with stateless JWT authentication and role claims enforced independently at both the routing layer and every API endpoint.
- Designed a normalized MySQL schema (11 tables, 1200+ seeded rooms) with foreign-key integrity across students, rooms, maintenance tickets, staff, and room-change requests.
- Implemented an end-to-end maintenance ticketing workflow (raise, triage, assign category-matched staff, resolve), backed by a REST API of 25+ endpoints across six resource controllers.
- Diagnosed and fixed a silent data-integrity bug where the UI displayed suggested staff contacts without ever persisting the assignment, leaving every ticket unassigned in the database despite appearing handled.
- Deployed a decoupled architecture (Render + Railway) with environment-driven API configuration, allowing the same frontend build to run against local and production backends.

AlgoMind — RAG-Based DSA Knowledge Assistant

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, ChromaDB, and Sentence Transformers for semantic search and contextual retrieval across a knowledge base of 100+ DSA topics.
- Engineered the ingestion path — chunking source material with overlap, generating embeddings via Sentence Transformers, and persisting them to a ChromaDB vector store supporting incremental indexing.
- Designed the retrieval and prompt-assembly layer to rank top-*k* passages and inject them as grounded context, anchoring answers to the indexed corpus rather than unverified model output.
- Implemented fault-tolerant multi-LLM fallback logic across Gemini and OpenAI APIs, ensuring 99%+ uptime and reliable responses under service failures.

SKILLS

Languages: Python, JavaScript, SQL

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming

Frameworks & Tools: React, Node.js, Express, Git/GitHub

Databases: MySQL, MongoDB

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

ACHIEVEMENTS AND CERTIFICATIONS

- MongoDB Certified Associate Developer — Python (Issued by MongoDB)
- Salesforce AI Associate Certification (Issued by Salesforce)
- Infosys Springboard Certification — Python (Jun 2024)
- Ranked in Top 3% in TS EAMCET (2023) among 200,000+ candidates

LEADERSHIP AND EXTRACURRICULARS

- Organising Committee, CBIT eSports Club** — ran inter-college tournaments with 500+ participants; led a 20-member team.
- Volunteer, CBIT Tech Fest** — ran coding workshops for 50+ participants on web development and AI fundamentals (Oct 2024).